

THE IMPACT OF PUBLIC TRANSFER EXPENDITURES ON TAX EVASION

An Experimental Approach*

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This paper outlines the methodology and results of an experiment undertaken to analyze, primarily, the impact of public sector transfer payments on tax-evasion behaviour. Apart from the use of transfer payments, further explanatory variables used were income, expected auditing probability and perceived tax burden. The influence of these variables on both the propensity to evade taxes and the extent of taxes evaded are investigated. In an attempt to add more 'realism', some innovative changes to the experimental approach were incorporated.

1. Introduction

Experimental approaches dealing with tax-evasion behaviour are normally based on micromodels, in the tradition of Allingham and Sandmo (1972). These models assume a utility function for the taxpayer which depends, inter alia, on after-tax income. After-tax income can, of course, be influenced by underreporting. The impact of tax rate, auditing probability and magnitude of fine on the maximization of expected utility, and therefore on the undeclared amount of income, is the main focus of analysis in these models.

In experimental approaches to tax evasion, the participants are faced with a 'game-simulation' situation in which they have the opportunity to evade

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taxes. This approach was used by Friedland, Maital and Rutenberg (1978) who analyzed the influence of tax rate, auditing probability, magnitude of fine and socioeconomic factors (age, gender, marital status etc.) on tax-evasion behaviour. Spicer and Thomas (1982) and Friedland (1982) examined the impact of the quality of information about the auditing probability and the penalty rate. Spicer and Becker (1980) extended the analysis by examining the relationship between 'fiscal inequity' and tax evasion. They tested whether or not the participants, who perceived their personal tax burden as higher than that of the other individuals, would evade more taxes. The influence of 'fiscal inequity' was also analyzed within the framework of a business simulation experiment by Webley, Morris and Amstutz (1985) who also examined the impact of the taxpayer's beliefs about the after-tax income of others. Spicer and Hero (1985) dealt with the influence of experience with tax evasion as well as with its detection, and the levels of evasion by other people, on the amount of taxes evaded.

In the experiment reported in this paper the analysis is further extended, by taking some aspects of public expenditure into consideration. Furthermore, the impact of perceived tax burden, expected auditing probability and income were also analyzed. Section 2 briefly outlines the hypotheses concerning the four mentioned determinants, section 3 describes the basic experiment, the results are provided in section 4, and section 5 concludes the article.

2. The hypotheses

Mackscheidt (1984, pp. 155–156) argues that taxpayers do not perceive their tax burdens as isolated sacrifices. Rather, the perception of the personal tax burden depends both on the amount of taxes paid by others and on the personal benefit derived from public expenditures. While the impact of the former determinant was analyzed by Spicer and Becker (1980) and Webley, Morris and Amstutz (1985), the present study examines the impact of the latter determinant.

Since it is rather complicated to simulate experimentally the impact of all types of public expenditure, it was decided to consider only *transfer payments*. It is hypothesized that taxpayers compare their received transfer payment with those of others and, if they see that they receive less, they will be more inclined to evade taxes. So, it is expected that personal transfer payments will be negatively correlated with the amount of taxes evaded.

While personal transfer payments influence tax evasion indirectly, the *perceived tax burden* affects the latter directly. If a taxpayer feels that his/her tax burden is high in relation to his/her received transfer payment, it is expected that he/she evades more taxes than individuals who do not feel so [Lewis (1982, p. 129)].

The third explanatory variable mentioned above, is the *expected auditing probability*. In reality, individuals do not know the auditing probability of their income declaration. However, every individual, who decides to evade taxes, expects a certain detection probability or frequency, which influences declaration behaviour. It can be assumed that the expected auditing probability has a negative impact on tax evasion.

The last determinant, which is taken into consideration, is *income*. Micro-models of tax evasion normally assume that an individual's risk aversion decreases with increasing income and as such, income is expected to be positively correlated with the amount of taxes evaded [Pencavel (1979, p. 116)].

3. The experiment

The experimental design was based on that developed by Güth and Mackscheidt (1985). They were the first to take public transfer expenditures into consideration. Furthermore, a progressive tax-rate table existed in their game-simulation situation. Transfers depended on the *actual* total revenue of taxes and fines, and were paid according to transfer coefficients individually assigned. The auditing probability, 0.17, was communicated to the participants.

In the game-simulation for this study, which also contained public transfer expenditures and the same progressive tax-rate table, transfers depended on the *expected* sum of all tax payments. In this way it was intended to better simulate reality, since in practice transfer amounts are fixed at the beginning of a period and are based on expected public revenues. The second difference to the experimental design developed by Güth and Mackscheidt was the uncertainty concerning the auditing probability: as in reality, auditing probability was not known to the participants. Third, in contrast to Güth and Mackscheidt, who assigned income for which they expected no service in return, it was attempted to simulate somewhat a real 'job' situation in which the participants had to 'work' for their income.

Two identical experiments were carried out, in the Universities of Bonn and Cologne respectively: 85 students of economics participated in Bonn, 31 in Cologne. The students were told that they had the opportunity to earn income which they had later to declare. The tax on their income would depend on their income declaration. The participants were told that they would receive transfers which were independent of the amount of their personal income. Income declarations were audited at random and the students were not informed of the auditing probability. If tax evasion was detected, i.e. if real income exceeded the declared income, the tax evader had to pay a penalty.

The level of income earned by each participant was based on a test, in

which he/she was confronted with 25 numerical series. These numerical series consisted of at least three and at most eleven numbers following certain numerical patterns. The student's task was to find these numerical patterns and to complete the numerical series in the correct order. After doing the test, income based on the individual's test result was paid to the students.¹

Three rates (t_i) of income taxes were fixed, for the income brackets given in table 1. Incomes (y_i) not falling in the fixed intervals did not exist, by design, and were not allowed to be declared.

The participants were told that the total amount of transfers would correspond to the expected sum of all tax payments – which was unknown to them. In order to give the participants the opportunity of comparing their transfer payments with those of others, they were informed of their individual *share* of total transfer payments, but they did not know the amount. In this way it was attempted to simulate reality, in that the taxpayer does not know the exact amount of all transfer payments received. Only three different transfer shares existed – 0.6%, 1.2% and 1.8% in Bonn, 1.7%, 3.4% and 5.1% in Cologne – and this was known to the students. In both experiments each percentage rate was assigned to almost the same number of participants. So the distribution of the transfer shares was as shown in table 2. The multiplication of each transfer share with the corresponding number of students, and the addition of these three products do not yield 100%. The differences can be attributed to the fact that in both experiments the percentage rates were rounded for simplification.

The expected sum of all tax payments was 300 DM in Bonn and 106 DM in Cologne.² On the basis of this, three *levels* of transfer payments were actually given – 1.80 DM, 3.60 DM and 5.40 DM respectively, corresponding

Table 1

y_i	t_i
$y_i < 4.50 \text{ DM}$	$\frac{1}{3}$
$6 \text{ DM} < y_i < 10 \text{ DM}$	$\frac{2}{3}$
$y_i < 13 \text{ DM}$	$\frac{1}{3}$

¹Only the following income amounts were possible: 2, 3, 4, 7, 8, 9, 14, 15 and 16 DM.

²These figures were based on the assumption that each possible income amount would be earned by almost the same number of students. Moreover, it was assumed that, on an average 25% of the income tax would be evaded. This implied the following expected sums of all tax payments:

Bonn: $(3 \text{ DM} \times 28 \times \frac{1}{3} + 8 \text{ DM} \times 29 \times \frac{1}{3} + 15 \text{ DM} \times 28 \times \frac{1}{3}) \times 0.75 = 297 \text{ DM}$;

Cologne: $(3 \text{ DM} \times 10 \times \frac{1}{3} + 8 \text{ DM} \times 11 \times \frac{1}{3} + 15 \text{ DM} \times 10 \times \frac{1}{3}) \times 0.75 = 108 \text{ DM}$.

Instead of these exact sums the amounts 300 DM and 106 DM were communicated to the students so that the individual transfer shares (in percentage rates) corresponded to the same absolute transfer amounts in both experiments.

Table 2

Bonn		Cologne	
Transfer share	Number of students	Transfer share	Number of students
0.6%	28	1.7%	10
1.2%	29	3.4%	11
1.8%	28	5.1%	10

to the respective transfer rating – and these were used as the transfer variables. The students in Bonn (Cologne) were told that these amounts would be given *if* the expected tax sum was 300 DM (106 DM in Cologne), but they were *not* told that the expected sum was 300 DM (106 DM).

Whether an income declaration of a participant was audited or not, was decided by throwing a dice. The auditing probability was 0.17 and was not known to the students.³ If tax evasion was detected, the tax evader had to pay a fine consisting of a fixed sum (2 DM) and a variable amount, the latter depending on the difference between real (received) income and declared income: the tax evader had to pay three times this difference multiplied by the tax rate on real income.⁴ It is important to note that the participants were told that, if the sum of tax and fine exceeded the sum of pre-tax income and transfer payment, they had to pay out of their own pocket. This requirement, it is felt, added considerably to the ‘realism’ of the experiment.

During the experiments the participants were asked for the expected auditing probability. Furthermore, they had to answer the question whether they regarded their tax burden – in comparison with their transfer payment – as too low, fair or too high. This question was asked in order to measure the variable perceived tax burden, and thereby to test the hypothesis mentioned in section 2. Since the transfer payment served as a standard of comparison, though, it may be the case that this measure of perceived tax burden is too similar to the variable expected net transfer.

4. Results

As in other studies [Friedland, Maital and Rutenberg (1978), Spicer and Thomas (1982)], the results concerning the influence of the explanatory

³This does not mean that a sample of only 17% of all cases was used for statistical analysis. All participants – even those who had not been audited – had to hand over their income declaration forms, when the experiment was over.

⁴For simplification, the fine did not depend on the exact amount of taxes evaded, because that would require the consideration of two tax rates (on real income and declared income).

variables are presented with regard to the propensity to evade taxes (whether tax evasion occurred or not) and the extent of taxes evaded (if tax evasion occurred). For that reason the following section is divided in two parts: the first part deals with the question of whether or not the determinants income, expected auditing probability, transfer payment and perceived tax burden (compared with the transfer payment) influence the propensity to evade.

In 36 (31%) of all cases tax evasion occurred. These tax-evasion cases are analyzed in the second part, which examines whether the determinants mentioned above affect the extent of taxes evaded.

4.1. *The propensity to evade taxes*

Since the propensity to evade is a dichotomous dependent variable, it was decided to choose the method of logistic regression with the programme GLIM (Generalised Linear Interactive Modelling)⁵ for the statistical analysis. This method is based on the generalization of the usual linear models (e.g. regression analysis) and can be used for qualitative variables. The dependent variable follows the binomial distribution, and the probabilities of tax evasion are transformed to logits. The coefficients are estimated by means of the maximum likelihood procedure. The results are as shown in table 3.

Table 3

<i>Dependent variable:</i>		
Propensity to evade taxes: $\log \frac{e}{1-e}$		
(e is the probability of tax evasion.)		
	Estimate	Standard error
Constant	0.761	0.944
<i>Independent variables:</i>		
Income	0.092 ^b	0.044
Expected auditing probability	-0.028 ^a	0.015
Transfer payment	-0.367 ^a	0.194
Perceived tax burden	-1.266 ^b	0.557
(too low/fair = 0; too high = 1)		

^{a, b}Indicate significance at 0.10 and 0.05 level of confidence (two-tail) respectively.

log-likelihood ratio statistic = 128.9; $df = 109$ ($p = 0.092$)

73.7% of all cases can be predicted correctly by the model.

⁵For the theoretical foundation of this method, see McCullagh and Nelder (1983) and Baker and Nelder (1978).

With the exception of the perceived tax burden the estimated coefficients of all variables have the expected sign: the propensity to evade taxes

- rises with increasing income,
- is negatively correlated with expected auditing probability,
- and decreases with rising transfer payments.

Surprisingly, the estimate of the variable perceived tax burden has a negative sign: participants, who regarded their tax burden in comparison with their transfer payment as too high, were less inclined to evade taxes than others. Since this result had not been expected, the possibility of multicollinearity between *perceived* tax burden and income was explored. The *real* tax burden is positively correlated with income according to the progressive tax-rate table and hence it could be assumed that perceived tax burden increases with rising income. However, the calculated Spearman rank-correlation coefficient (r_s) is very low (0.097) and not significant.

As the participants had to rate their tax burden in comparison with their transfer payment, it could be argued that the unexpected result is due to the influence of the transfer payment. This varies inversely with the ratio of real tax burden to transfer payment and influences therefore the perceived tax burden negatively.⁶ So, the latter would have a negative impact on the propensity to evade taxes, if this is positively correlated with the transfer payment. That would be the case if the participants perceived the transfer payment as a part of income, which exercises a positive influence on the propensity to evade taxes. However, this income effect cannot be observed, since the estimate of the variable transfer payment has a negative sign.

Perhaps the negative impact of the perceived tax burden can be explained on the assumption that individuals, who perceive high tax burdens, assess their after-tax income as too low to be able to pay the fine in the case of detection. This could be accompanied by an increase of risk aversion, and and it may be, therefore, that they have only a slight propensity to evade taxes.

4.2. *The extent of taxes evaded*

In this part the behaviour of only the tax evaders (36 of all cases) is analyzed. The only variable, which significantly influences the dependent variable, percentage of taxes evaded, is expected auditing probability: according to a priori expectation the extent of taxes evaded is negatively correlated with expected auditing probability ($r_s = -0.476$; $p < 0.01$). However, with regard to the other variables – income, transfer payment and perceived tax burden – no significant effect can be observed.

⁶The correlation coefficient r_s (for transfer payment and perceived tax burden) is -0.371 ($p < 0.01$).

The phenomenon that explanatory variables do affect the propensity to evade taxes but, with the exception of expected auditing probability, not the extent of taxes evaded can occur, if many of the participants were at corner solutions. In this case participants, who had decided to evade, would have declared that income amount which maximized their expected after-tax income. In order to examine this, for each participant the optimum percentage of taxes evaded, i.e. the proportion of taxes evaded which maximizes – with regard to the participant's expected auditing probability and real income – the after-tax income, was calculated and compared with the corresponding actual percentage.⁷ The optimum percentage was evaded in only six of 36 cases,⁸ but there exists a significant positive correlation between optimum and actual percentage ($r_s = 0.465$; $p < 0.01$). So, it is likely that, although few tax evaders were at corner solutions, participants *tried* to maximize their after-tax income. This would explain that the extent of tax evasion was influenced by expected auditing probability, since only this explanatory variable is (negatively) correlated with optimum percentage of taxes evaded ($r_s = -0.899$; $p < 0.01$).⁹

5. Conclusion

As Spicer and Becker (1980, pp. 174–175) and Spicer and Thomas (1982, p. 241) have pointed out, the results of experimental approaches must be interpreted cautiously. The main reason is the artificiality of the game-simulation situation which might induce the participants to act differently than in reality. For example, the effect of stigmatization, under which a detected tax evader may suffer in reality, was not considered. Although it may be possible to simulate such a deterrent effect,¹⁰ it was not considered in the experiment described above which appeared already to be complicated enough. On the other hand, the experimental approach has the advantage in that it allows direct observation of behaviour that may be penalized: the results depend not on the honesty of responses as in survey research. Furthermore, isolated influences of explanatory variables can be better examined in a laboratory setting.

⁷Although two cases with two optimum percentages each were observed, the results presented below did not change significantly when one optimum percentage was substituted for the other in the computations.

⁸Concerning the group of participants, who did not evade taxes, this behaviour turned out to be optimal in 45 of 80 cases.

⁹In contrast to this there exists no significant correlation (0.049) between (real) income and optimum percentage.

¹⁰For example: participants could be told that those who have not evaded taxes determine the fine of a tax evader in the case of detection.

The results of the experiment reported here indicate that the propensity to evade taxes appears to depend on income, expected auditing probability, transfer payment and perceived tax burden. With the exception of perceived tax burden, the determinants influence the propensity to evade taxes according to a priori expectation. The variable perceived tax burden is the only one which has been analyzed in previous experiments [Spicer and Becker (1980), Webley, Morris and Amstutz (1985)]. However, in both experiments the explanatory variable was the perception of tax burden compared to that of others, not compared to transfer payment as was done here. Spicer and Becker examined the influence on only the dependent variable 'percentage of taxes evaded', and they found a significant positive impact [Spicer and Becker (1980, p. 174)]. As noted above, this variable was not affected significantly by perceived tax burden in the experiment reported here. Webley, Morris and Amstutz dealt with the influence on the dependent variables 'percentage of income declared' and 'frequency of under-declaring', but they found no significant effect [Webley, Morris and Amstutz (1985, p. 240)].

The main result of this study is that tax-evasion behaviour appears to depend not only on the public revenue and audit systems, but also on public expenditures. In the latter case, the individual transfer payment received from the public sector plays a decisive role. If a taxpayer supposes that he/she receives less than others, the probability that he/she evades taxes, rises. This means that a politician, who wants to use transfer payments as an instrument of economic policy, may need to consider carefully the impact of the transfer payments' distribution on tax-evasion behaviour.

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